TIPA Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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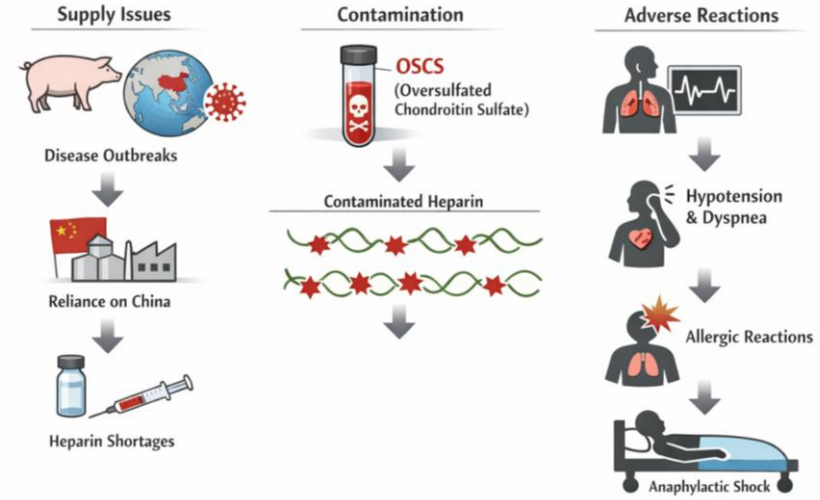
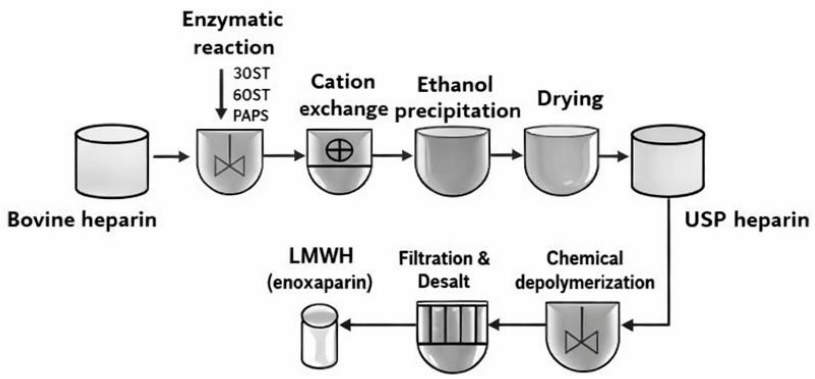
2. Technology Overview

Title of Proposed Technology	Development of a scalable enzymatic method for the production of USP heparin and low molecular weight heparin (LMWH) from bovine intestinal heparin			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Other Biopharmaceuticals	Industrial Biotechnology	Functional Biomaterials
Concept of Proposed Technology	Heparin is the oldest most frequently used anticoagulant in the world and is on the WHO's list of essential medicines. FDA-approved heparin is extracted from porcine intestines, with ~80% of active pharmaceutical ingredient (API) coming from China, with a worldwide market approaching $8B and over 120 metric tons/year. Bovine intestinal (BI) heparin is an existing alternative; however, it has much lower intrinsic anticoagulant activity than porcine heparin. We demonstrated that BI heparin activity could be increased to that of USP porcine heparin by treating with two sulfotransferases (3OST-1 and 6OST-3).[1-4] We also demonstrated conversion of this remodeled BI heparin to a low molecular weight heparin (LMWH) with identical properties to enoxaparin.[2] We now need to optimize and scale up the use of these sulfotransferases in a well-controlled process to generate sufficient bovine-derived heparin that is the equivalent of USP heparin at commercial scale. In addition, heparin is used as coatings on medical devices, and we will explore use of remodeled BI heparin in such a manner.			
Rationale for Joint R&D Project	Due to disease (i.e., swine fever, blue ear disease) and geopolitics, disruption of the heparin supply has occurred. Moreover, in 2007–2008 a major heparin contamination crises occurred as a result of purposeful contamination of porcine heparin from China with oversulfated chondroitin sulfate, resulting in nearly 100 deaths in the U.S. **Replacing porcine heparin with an alternative, more reliable, and safer source is a critical need to diversify the worldwide supply chain and eliminate chronic heparin shortages (Figure 1).** BI heparin is one such alternative. There is interest from regulatory agencies in broadening the source of heparin by through BI heparin. Also, some religious groups may refrain from using porcine-derived products, increasing the importance of a BI heparin. The basic technology has been developed, and we seek to develop efficient and cost-effective scale up with			

	industry partner(s) with interest in biocatalytic synthesis and scale-up.  **Figure 1.** Summary of potential problems with porcine sourced heparin.
Key Words	1. Heparin 2. Synthetic Biology 3. Enzymes 4. Anticoagulant 5. Biomaterial for Medical Devices

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (6)
R&D Objectives	We propose a pathway to eliminate the global chronic heparin shortages by designing and scaling up a biocatalytic pathway to modify BI heparin as raw material and endow it with properties of currently approved porcine heparin (**Figure 2**). We propose three tasks to advance to TRL 6:  **Figure 2.** Proposed schematic pathway for biocatalytic modification of BI heparin into USP-grade porcine-equivalent heparin. **Task 1:** Determine key structural features of a range of BI heparins and compare with USP porcine heparins and a fractionated BI heparin that has high anti-IIa and anti-Xa activities. **Deliverable:** Provide data on disaccharide composition, molecular weight, and anti-IIa and anti-Xa activities from a range of available BI heparins and compare to both the published range of USP porcine heparins			

and to a fractionated BI heparin with USP porcine heparin properties. **Go/No-Go**: Clear structural features of BI heparin pre- and post-fractionation are obtained and compared to USP porcine intestinal heparin. Prepare full proposal with Korean SME.

Task 2: Optimize the enzymatic production of enhanced BI using a scalable 3*O*ST-1 process and confirm biological equivalence (anti-IIa and anti-Xa activities) to USP porcine heparin. **Deliverable:** Produce at least 2 g of enhanced BI heparin using 3*O*ST-1 (and if required, 6*O*ST-3) and possessing biological properties similar to USP porcine heparin. Demonstrate conversion of enhanced BI heparin to enoxaparin.

Task 3: Perform pilot scale synthesis of 200 g enhanced BI heparin. **Deliverable**: An enhanced BI heparin produced at scale with key process variables identified and evaluated, and conversion of 100 g of enhanced BI heparin into enoxaparin.

Final Deliverable: Generate 200 g of BI heparin with chemical and pharmacological properties identical to USP porcine heparin and successful conversion to bovine heparin-derived enoxaparin (100 g).

R&D Target Specifications

The chemical and biological properties of the enhanced BI heparin/LMWH products should be matched with USP heparin/enoxaparin derived from porcine intestinal heparin[1, 2]:

	disaccharide composition (mol %)[a]										activity (U/mg)[b]		
heparin	0S	NS	6S	2S	NS6S	NS2S	2S6S	TriS	sulfo groups per disaccharide	Mw[c]	anti-Xa	anti-IIa	Xa/IIa
USP specs									2.5–2.7	15000–19000		>180	0.9–1.1
PI (USP) heparin	4.0	3.2	3.7	2.0	10.2	7.8	1.5	67.7	2.5	17000	200	200	1.0
BI heparin	3.3	5.9	0.8	2.5	8.2	32.2	0.2	46.8	2.3	18500	110	105	1.0
BI heparin 3S	3.9	6.3	0.0	2.5	5.7	32.1	0.0	49.5	2.3	18500	231	268	0.9
BI heparin 3S6S	3.6	5.4	0.0	2.4	6.1	15.3	0.0	67.2	2.5	17900	289	301	1.0

[a]The systematic errors of disaccharide compositional values are less than 0.1%. [b]Systematic errors of bioactivity assays are less than 5%. [c]Some of the larger chains may remain bound to the enzymes, reducing the average molecular weight of the resulting products.

	LMWH USP specifications							
	MW 3800–5000 Da	MW < 2000 Da 12–20%	MW 2000–8000 Da 68–82%	MW > 8000 Da NMT18%	1,6-anhydro 15–25%	anti-Xa (IU/mg) 90–125	anti-IIa (IU/mg) 20–35	anti-Xa/anti-IIa 2.5–5.5
Enoxaparin	4200	17	73	11	22	105	25	4.20

R&D Plan

The goal of this project is to develop a robust and scalable biocatalytic process for the enhancement of BI heparin into a heparin with properties essentially identical to USP Heparin Sodium from pig intestines. As a result, a new source of heparin can be achieved.

Background: Heparin is a highly sulfated glycosaminoglycan widely used as an anticoagulant, traditionally produced from animal tissues. For several decades, the global heparin supply has relied almost exclusively on porcine intestinal mucosa, which provides material that meets pharmacopeial requirements for anticoagulant potency, molecular weight distribution, and structural features critical for antithrombin binding. Historically, bovine-sourced heparin, derived from bovine lung or intestinal tissues, was also used clinically. However, bovine heparin differs structurally from porcine heparin, typically exhibiting lower overall sulfation, reduced content of the antithrombin-binding pentasaccharide, and altered disaccharide composition, resulting in lower anticoagulant potency per unit mass. In addition, concerns related to transmissible spongiform encephalopathies (TSEs) and batch-to-batch variability led to the withdrawal of bovine heparin from many markets in the late 1990s and early 2000s, leaving porcine sources as the dominant global supply.

Dependence on a single animal source has raised significant concerns regarding supply chain vulnerability, highlighted by the 2007–2008 heparin contamination crisis and by periodic disruptions in pig populations. Consequently, there has been renewed interest in reintroducing bovine heparin as a supplementary or alternative source, provided that safety, quality, and pharmacological equivalence can be demonstrated.

Bovine-sourced heparins exhibit different structures and specific activities (U/mg) than porcine intestinal heparins. The MW of bovine intestinal heparin is slightly lower than that of porcine heparin; it is also more polydisperse, less highly sulfated, and more heterogeneous.[5,6] Bovine intestinal heparin has also been shown to have lower content of GlcNS3S residues and a slightly higher content of GlcA residues than porcine intestinal heparin. There are also major differences in the biological activities of bovine-sourced and porcine-sourced heparins. Porcine intestinal heparin possesses significantly higher activity than bovine-sourced heparins (~180−210 U/mg and ~100−130 U/mg, respectively).[6]

Previous efforts to reestablish bovine heparin have largely focused on (i) Improved purification and fractionation methods to enrich high-activity components, (ii) Chemical or enzymatic modification strategies to increase sulfation levels and enhance anticoagulant activity, and (iii) Regulatory re-evaluation, including the development of separate pharmacopeial standards for bovine heparin in some regions. Notably, while these approaches have demonstrated that bovine heparin can be made clinically usable, most prior strategies have resulted in products that remain distinct from USP porcine heparin in fine structure, potency distribution, and biological profiles, often requiring separate dosing standards and specifications. More recently, interest has shifted toward biocatalytic and chemoenzymatic approaches that more closely mimic natural biosynthetic pathways, aiming to selectively modify sulfation patterns and epimerization states to achieve structures closer to pharmaceutical-grade porcine heparin. However, scalable processes capable of converting bovine intestinal heparin into material that is structurally and functionally equivalent to USP Heparin Sodium have not yet been established, and this remains a critical unmet need for diversifying the global heparin supply.

Preliminary Work: In last two decades, the studies from Rensselaer Polytechnic Institute (RPI), led by the Dordick and Linhardt groups, establish a comprehensive and scalable strategy to produce bioengineered heparin and low-molecular-weight heparin (LMWH) that are chemically and biologically comparable to current animal-derived products, while reducing reliance on porcine sources and improving supply security:

1). Enzymatic Remodeling of Animal-Derived Precursors (2017, J. Med. Chem.): Fu et al. (2017)[1, 3-4] demonstrated that bovine intestinal heparin, which naturally has lower anticoagulant activity than porcine heparin, can be enzymatically modified using two biosynthetic heparin enzymes (i.e. 3*O*ST-1 and 6*O*ST-3 sulfotransferases) to generate highly anticoagulant heparin with enhanced antithrombin binding and pharmacological activity. This work established the feasibility of post-extraction enzymatic upgrading of heparin to pharmaceutical-grade material.

2) Production of LMWH from Remodeled Heparin (2021, J. Med. Chem.): Baytas et al. (2021)[2] extended above approach by converting enzymatically remodeled bovine heparin into LMWH using controlled depolymerization. The resulting LMWH exhibited appropriate molecular weight distribution, anti-Xa/anti-IIa activity ratios, and structural features comparable to commercial LMWH products. This study demonstrated that both unfractionated heparin and LMWH can be produced from non-porcine starting materials through integrated enzymatic and chemical processing.

3) Full Bioengineered Heparin Platform and Conversion to LMWH (2024, PNAS): Douaisi et al. (2024)[7,8] reported a hybrid biological-chemical platform that synthesizes heparin starting from microbially produced heparosan, followed by stepwise enzymatic sulfation, epimerization, and chemical finishing to yield bioengineered heparin closely matching porcine heparin in structure, anticoagulant activity, and pharmacological performance. Importantly, the product could be readily converted to LMWH using standard industrial methods, demonstrating compatibility with existing manufacturing pipelines and regulatory frameworks.

Together, these papers demonstrate a modular, enzyme-enabled manufacturing paradigm for bioengineered heparin and LMWH that (i) Reduces dependence on porcine intestinal mucosa supply chains; (ii) Enables use of alternative animal or microbial polysaccharide precursors; (iii) Produces materials with pharmaceutical equivalence to current commercial products; (iv) Is compatible with current industrial LMWH production processes; and (v) Supports scalable, controllable, and safer heparin manufacturing. In addition, we have significant experience in developing a large-scale biosynthetic heparin process based on a chemo-enzymatic process and heparin related analytical works.[5-15] These comprehensive studies position RPI as a global leader in developing next-generation, supply-secure, bioengineered anticoagulants with direct relevance to regulatory-grade manufacturing.

Task 1: (Months 0–6; TRL 4). Determine key structural features of a range of BI heparins and compare with USP porcine heparins and a fractionated BI heparin that has high anti-IIa and anti-Xa activities. Specific subtasks are as follows:
Task 1a: Obtain a number of BI heparins and determine ranges of disaccharide composition, molecular weight, and anti-IIa and anti-Xa activities (similar to that shown in Table 1).
Task 1b: Use ion-exchange chromatography to fractionate[16] and isolate BI heparin chains with disaccharide composition, molecular weight, and anti-IIa and anti-Xa activities consistent with USP

porcine heparin and determine yields of this fractionation.
Disaccharide Compositional Analysis: We will perform comprehensive disaccharide compositional analysis to characterize the structural features of heparin samples as per our previous publication.[11] Authentic disaccharide standards will be used for peak identification and quantification.
Molecular Weight Determination: The molecular weight distributions of heparin samples will be determined by gel permeation chromatography (GPC) essentially following the USP Heparin monograph (USP 37) method.[11] Number-average (Mn), weight-average (Mw), and polydispersity index (PDI) will be determined.
***In Vitro* Anticoagulant Activity Measurement:** The anticoagulant activities of heparins will be evaluated by measuring anti–factor Xa (anti-Xa) and anti–factor IIa (anti-IIa) activities using validated two-stage chromogenic assays (BIOPHEN Heparin Anti-Xa and Anti-IIa kits, Aniara). These assays quantify heparin-dependent inhibition of factor Xa or thrombin in the presence of antithrombin III (ATIII). Anti-Xa and anti-IIa activities will be calculated using calibration curves generated with USP heparin standards (0–1 U/mL) and expressed as units per milligram of heparin. These measurements will be used to compare anticoagulant potency and anti-Xa/anti-IIa ratios among heparin preparations and to correlate bioactivity with structural features determined by chromatographic and compositional analyses.
Deliverable: Prepare full proposal with Korean SME.
Milestones: Data provided on disaccharide composition, molecular weight, and anti-IIa and anti-Xa activities from a range of available BI heparins, and compared to both the published range of USP porcine heparins and to a fractionated BI heparin with USP porcine heparin properties.

Task 2: (Months 7–24; TRL 5). Optimize the enzymatic production of enhanced BI using a scalable 3*O*ST-1 process (Figure 3) and confirm chemical and biological equivalence (anti-IIa and anti-Xa activities) to USP porcine heparin. An enzyme-based method will be used to enhance the anti-coagulant activity of BI heparin to levels equivalent to USP porcine heparin. 3*O*ST-1 and 6*O*ST-3, and the sulfo-donor, 3′-phosphoadenosine-5′-phosphosulfate (PAPS), will be used to increase TriS and 3-*O*-sulfation levels of BI heparin, creating additional ATIII binding sites, thus enhancing anticoagulant activity. Both enzymes have been generated recombinantly in *E. coli* in our lab[13,14] containing a maltose binding domain (6*O*ST-MBD) or His tag (His_6-3*O*ST), followed by immobilizing onto several supports. We envision that using biosynthetic enzymes in a well-controlled process will afford an enhanced BI heparin that is the generic equivalent of USP porcine heparin in a cost effective process at commercial scale. Specific subtasks are as follows:
Task 2a: Produce 1 gram of recombinant 3*O*ST-1 and 6*O*ST from *E. coli* and generate an enhanced BI heparin with anti-IIa and anti-Xa activities consistent with USP porcine heparin. Initial work will involve a soluble enzyme process with enzyme plus the sulfo donor PAPS, the latter at concentrations consistent with the conversion of 3-*O*-sulfate groups required for enhanced BI heparin activity.
Task 2b: Assess the ability of recombinant 3*O*ST-1 produced in *E. coli* to convert a crude BI heparin into enhanced BI heparin. These experiments will demonstrate that crude heparin does not affect 3*O*ST-1 activity.
Task 2c: Immobilize recombinant 3*O*ST-1 (from *E. coli*) and generate an enhanced BI heparin with properties as specified in Task 2a. Commercially relevant supports will be used, including agarose, methacrylate-based and silica, and various immobilization chemistries will be examined, including cyanogen bromide, epoxy and affinity-based. Employ either a stirred-tank reactor or a packed-bed reactor with enzyme bound to the optimal resin. The immobilized enzyme will enable reuse of the biocatalyst and the stability of the enzyme for reuse will be determined using colorimetric assays. Use the optimal reactor configuration and enzyme immobilization strategy to generate 2 g of enhanced BI heparin with properties as specified in Task 2a.
Task 2d Immobilize 6*O*ST-3 as with 3*O*ST-1 in Task 2d, and generate enhanced BI heparin from BI heparin at 50 mg scale. Immobilized 6*O*ST-3 will be used in a first reaction and immobilized 3*O*ST-1 (Task 2c) will be used in a second reaction. Analyze composition of the resulting enhanced BI heparin (i.e., disaccharide composition) and properties as specified in Task 2a.
Task 2e: Convert enhanced BI heparin to enoxaparin using the chemical process from our previous work (**Figure 4**).[2] Specifically, convert 100 mg of enhanced BI heparin to enoxaparin using literature procedures. Determine composition, molecular weight, activity of the generated enoxaparin and reaction yield. Compare chemical and biological properties to USP enoxaparin derived from porcine intestinal heparin. Measure endotoxin levels in the BI-derived enoxaparin.
Deliverable: Produce at least 2 g of enhanced BI heparin using 3*O*ST-1 (and if required, 6*O*ST-3) and possessing biological properties similar to USP porcine heparin. Demonstrate conversion of enhanced BI heparin to enoxaparin.

Task 3: (Months 25–36; TRL 6). Perform larger pilot scale synthesis of 200 g of enhanced BI heparin at RPI and conversion of 100 g into a LMWH. Specific subtasks are as follows:

Task 3a: Produce 50 g of recombinant 3*O*ST-1 enzyme (and 6*O*ST-3 if needed) from *E. coli*. Use immobilized enzymes and PAPS recycling (if 6*O*ST-3 is required), starting with 10 g of crude BI heparin, to generate enhanced BI heparin. Determine composition, molecular weight, activity of the generated enhanced BI heparin, and reaction yield. If 6*O*ST-3 is used, determine the disaccharide composition. Evaluate reuse of enzyme(s) by determining composition, molecular weight, and activities of the generated enhanced BI heparin. This task will enable multiple 10 g runs to be performed leading to an optimized small-scale process ready to be used on Task 3b.

Task 3b: Scale up to 200 g feed of crude BI heparin using immobilized enzyme reactors.

Task 3c: Convert 100 g of enhanced BI heparin to enoxaparin using literature procedures[2] and measure yield. Compare chemical and biological properties to USP enoxaparin derived from porcine intestinal heparin.

Deliverables: An enhanced BI heparin (200 g) produced at scale with key process variables identified and evaluated, and conversion of 100 g of enhanced BI heparin into enoxaparin (LMWH).

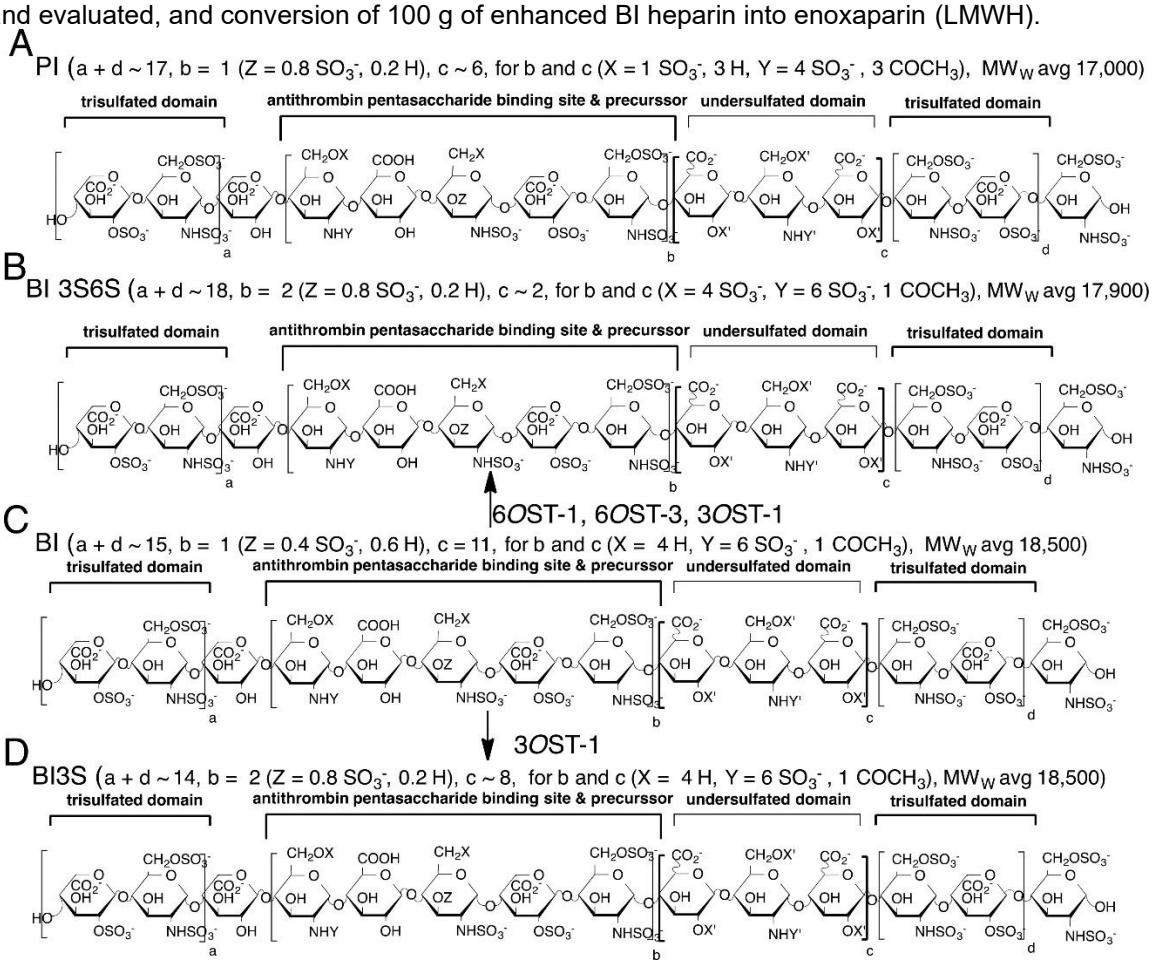
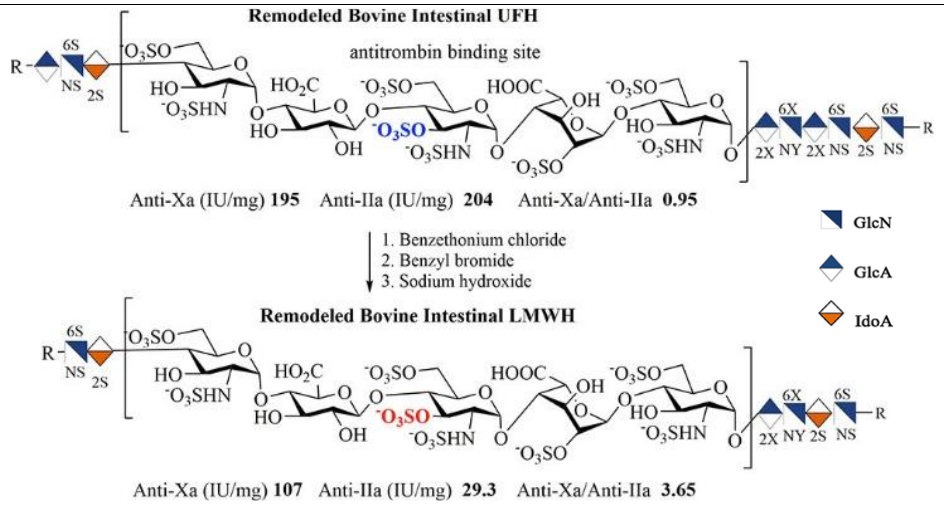


Figure 3. Structure of heparins and their remodeled structures. a + d ~ 20, b = 1, c ~ 8 for MW weight average 20 000. X or X′ = SO_3^-, H; Y or Y′ = SO_3^-, $COCH_3$, H; Z = SO_3^-, H. (A) Schematic structure of an average chain present in PI heparin. (B) Schematic structure of an average chain present in BI 3S6S heparin. (C) Schematic structure of an average chain present in BI heparin. (D) Schematic structure of an average chain present in BI 3S heparin.[1]

	 **Figure 4.** Preparation of LMWH from remodeled bovine intestinal heparin. Remodeled bovine intestinal heparin (X = 4H; Y = 6 SO_3^-, 1 $COCH_3$) shown in combined chemical and symbolic structures.[2]

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Research Institute will serve as a technical partner providing specialized expertise in recombinant enzyme expression/purification, heparin-related enzymology and glycan analysis to support development of a scalable enzymatic process for producing USP-grade heparin and LMWH from bovine intestinal heparin. The Institute will perform structural and functional characterization of process intermediates and final products, including molecular weight and polydispersity analysis (GPC), disaccharide composition (LCMS), and in vitro anticoagulant activity assays (anti-Xa and anti-IIa) in accordance with USP and pharmacopeial methods. These data will guide process optimization and confirm product comparability to USP reference standards. All tasks will be part of the RPI effort in support of larger-scale enzyme production and enhanced BI heparin production by the Korean SME. **Team** **Jonathan S. Dordick, Ph.D.,** is Institute Professor of Chemical and Biological Engineering at RPI, with a joint appointment in the Department of Biological Sciences. He also serves as Vice President for Strategic Alliances and Translation. He received his B.A. degree in Biochemistry and Chemistry from Brandeis University and his Ph.D. in Biochemical Engineering from the Massachusetts Institute of Technology. At RPI, he has served as the Vice President for Research, founding Co-Director of the Rensselaer-Mount Sinai Center for Engineering and Precision Medicine, Director of the Center for Biotechnology & Interdisciplinary Studies, and Department Chair. Dr. Dordick has made foundational contributions to protein science and enzyme technology, biomanufacturing of therapeutic proteins and nucleic acids, and drug discovery. Dr. Dordick has received numerous awards and honors, including election to the National Academy of Engineering and National Academy of Inventors, and receipt of numerous society awards. He has founded several companies, and published over 460 papers and is an inventor/co-inventor on over 50 patents. **Fuming Zhang, Ph.D.,** is a Research Professor/Senior Research Scientist in the Department of Chemical and Biological Engineering at RPI. Dr. Zhang has a broad background in biophysics and glycobiology, with specialized training and expertise in key research areas relevant to this proposal. As PI or Co-Investigator on several previous NIH-funded grants, he established the groundwork for the developing robust protocols for glycan purification and structural characterization using NMR and LC-MS, as well as methods for analyzing GAG–protein and protein–protein interactions using SPR. Dr. Zhang published over 400 peer-reviewed papers and book chapters and hold several patents in the areas of biochemistry, glycoscience and glycoengineering.
Expected R&R of Korean SME Partner	The Korean SME will serve as the industrial development partner responsible for process optimization, scale-up, and pilot-scale validation of the enzymatic production of USP-grade heparin and LMWH from bovine intestinal heparin, and will be engaged in all tasks. The SME will develop scalable reaction and downstream purification processes, establish process controls, and evaluate yield, reproducibility, and cost-effectiveness under manufacturing-relevant conditions. We expect SME has the expertise in enzyme scale-up/production, biocatalysis, and heparin manufacturing. The

	SME will conduct pilot production runs and provide materials for independent analytical testing, while performing in-house quality assessments of critical attributes such as molecular weight distribution and impurity profiles. Resource commitments will include pilot-scale equipment, technical staff, and project management support. The SME will collaborate closely with academic and research partners to incorporate analytical feedback into iterative process improvements and to support technology transfer and commercialization planning.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This project is expected to deliver a validated, scalable enzymatic process for converting BI heparin into USP-compliant unfractionated heparin (an enhanced BI heparin) and clinically relevant low molecular weight heparin (LMWH) derived from enhanced BI heparin. Key outcomes will include the following: • Optimized enzyme-based production protocols. • Defined critical process parameters. • Well-characterized products that meet pharmacopeial standards for molecular weight distribution, structural composition, purity, and anticoagulant activity. • Robust analytical and quality assessment workflows that link process conditions to critical quality attributes (CQAs), enabling consistent and reproducible production suitable for future GMP implementation. • Pilot-scale validation will demonstrate feasibility for industrial manufacturing and provide a clear pathway toward regulatory translation. The impact of this work will be significant for both public health and pharmaceutical manufacturing. By enabling a non-porcine, enzyme-driven production route, the technology will diversify the heparin supply chain, reduce vulnerability to raw material shortages and contamination risks, and enhance drug safety. Collectively, the outcomes will strengthen supply security, improve manufacturing sustainability, and facilitate broader access to essential anticoagulant therapies. The scalable enzymatic platform may also be adapted for the production of structurally defined heparins, next-generation anticoagulants, and non-anticoagulant heparin-derived (bio)materials. With respect to the latter, heparin is a highly negatively charged polysaccharide that plays essential roles in numerous physiological and pathological processes through its interactions with proteins. These interactions regulate cellular signaling, inflammation, coagulation, and host–pathogen recognition. For example, interactions between viral capsids and cell-surface sulfated glycans (such as heparin and heparan sulfate) can serve as early attachment factors in viral pathogenesis. In our laboratory, we have discovered that heparin and other glycomimetics, including marine sulfated glycans, exhibit antiviral activity and other bioactivities [17-18].

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization